ACADEMIC PROJECT EVALUATION REPORT

Department of Computer Science & Engineering / Information Science
Continuous Assessment & Evaluation

CAMPUSCONNECT AI

Generative AI Powered Student Assistance Chatbot

UNITED NATIONS SUSTAINABLE DEVELOPMENT GOAL 4: QUALITY EDUCATION

Project aligned with United Nations Sustainable Development Goal 4 (Quality Education): Ensuring inclusive and equitable quality higher education and promoting lifelong learning opportunities through 24/7 accessible, grounded Generative AI guidance.

Academic Submission Details

Project Parameter	Official Specification
Author Designation	Student Author / Project Contributor
Institution	Engineering Institution / University
Project Title	CampusConnect AI – Generative AI Powered Student Assistance Chatbot
Selected United Nations SDG	SDG 4 – Quality Education (Targets 4.3, 4.4, 4.a)
Academic Year	2026
Frontend Architecture	HTML5 (Semantic) • CSS3 (Navy & Electric Blue) • Vanilla JS (ES6+)
Conversational AI Engine	Botpress Cloud (Autonomous Node, NLU, RAG Vector Engine)
Open Source GitHub Repository	https://github.com/Immortal-syntax/CampusConnect
Live Production Deployment	https://campus-connect-kohl-alpha.vercel.app/
Operational Status	Completed, Audited, Open Source & Deployed (HTTP 200)

1. Selected SDG and Reason for Selection

1.1 Overview of United Nations SDG 4 (Quality Education)

In September 2015, the United Nations General Assembly adopted the 2030 Agenda for Sustainable Development, comprising 17 Sustainable Development Goals (SDGs). Among these, Sustainable Development Goal 4 (SDG 4) is formally defined as:

> **United Nations SDG 4 Mission Statement**
> *"Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all."*

SDG 4 recognizes that high-quality education is the fundamental catalyst for enabling upward socioeconomic mobility, reducing inequality, and unlocking individual human potential. While primary education establishes foundational literacy, higher tertiary and technical education equip students with specialized cognitive, computational, and problem-solving competencies essential for modern industrial and knowledge economies.

1.2 Importance of Equal Access to Quality Education

Equal access to educational support ensures that academic success is determined by intellectual merit rather than socioeconomic privilege or administrative familiarity. When academic guidance is confined to physical offices during limited 9 AM to 5 PM business hours, commuter, working, and first-generation learners face severe informational handicaps. Equalizing access through always-on conversational guidance directly dismantles these structural barriers.

1.3 Challenges in Accessing Academic Guidance

Higher education students routinely encounter systemic obstacles when attempting to obtain academic guidance:

• Severe Information Fragmentation: Guidelines, exam rules, and student services are dispersed across siloed websites and static PDFs.

• Administrative Queuing Latency: Enrollment weeks and exams result in multi-day backlogs for academic advising desks.

• Out-of-Hours Guidance Void: Students study and experience academic anxiety during evening and late-night hours when advising desks are closed.

• Resource Discovery Friction: Many students remain unaware of accredited, government-funded open learning repositories (SWAYAM, NPTEL, NDLI).

1.4 Relevance of CampusConnect AI to SDG 4

SDG 4 Target	International Mandate	CampusConnect AI Direct Operationalization
Target 4.3	Equal access to affordable tertiary and technical education	Provides 24/7 continuous guidance on degree requirements, course add/drop deadlines, and prerequisite rules, ensuring commuter and working students have identical access to advising.
Target 4.4	Increase youth competencies and digital skills	Delivers an ultra-lightweight, high-performance web application built with native web standards (HTML5/CSS3/Vanilla JS) that operates flawlessly on

		low-cost smartphones and weak networks.
Target 4.a	Build and upgrade inclusive and safe learning facilities	Instantly directs students to campus psychological counseling (CAPS), disability accommodations, peer tutoring centers, and emergency student financial relief funds.

2. Problem Statement

In higher educational institutions, students must navigate a highly intricate operational environment governed by extensive academic regulations, prerequisite trees, examination ordinances, and student life policies. However, the systems used to communicate this information to students are fundamentally manual, fragmented, and latency-prone.

2.1 Operational Context & Affected Stakeholders

This problem is prevalent across engineering colleges, universities, and technical institutes where administrative data is divided among isolated silos: the Registrar manages registration; the Examination Controller manages exam timetables and GPA scales; academic departments oversee prerequisites; and Student Affairs oversees hostels and counseling. Affected stakeholders include:
1. Undergraduate & Graduate Students: Experiencing advising delays, missed add/drop deadlines, and severe exam anxiety.
2. Faculty Advisors: Overburdened by repetitive transactional queries (e.g. 'How do I drop a class?').
3. Administrative Staff: Facing overwhelming communication surges during semester milestones.

2.2 Limitations of Status Quo Channels

Physical advising desks operate strictly during business hours, queues require physical presence, and institutional circulars are often published as non-searchable scanned PDFs. Left unaddressed, academic probation rates increase, student retention suffers, and administrative costs needlessly escalate.

3. Proposed Solution

CampusConnect AI pairs an accessible, dependency-free web portal with an authentic Botpress Cloud Generative AI assistant powered by Retrieval-Augmented Generation (RAG). The platform acts as a 24/7 digital academic advisor that answers student inquiries in natural language, grounding its outputs strictly in verified institutional documents.

3.1 System Architecture & Data Flow

```
Student Client (HTML5 / CSS3 / Vanilla JS)
  │
  ▼  (Official Webchat v3.7 Embed)
Botpress Cloud NLU & Autonomous Node
  │
  ▼  (Vector Similarity Search)
RAG Knowledge Base (/knowledge-base/*.md)
  │
  ▼  (Grounded Context Injection)
```

```
LLM Synthesis & Anti-Hallucination Guardrail
  |
  ▼  (Real-Time Streaming Response)
Student Webchat Interface
```

3.2 Anti-Hallucination Guardrails

To protect academic integrity, the agent enforces strict behavioral boundaries:

• Refusal to Guess Dates/Fees: Explicitly directs students to official college portals for semester dates, tuition, and cutoffs.

• Academic Integrity: Refuses to write complete graded assignments; instead explains concepts and outlines logic.

• Scope Enforcement: Politely redirects non-academic queries back to collegiate guidance.

4. Project Features

The live production release incorporates 12 fully verified capabilities:

1. Responsive Student Portal: Accessible Navy/Electric Blue SaaS layout fluid across 320px to 4K displays.
2. Header Navigation & Mobile Drawer: Keyboard-accessible anchor navigation with animated mobile drawer.
3. UN SDG Banner & Live Indicator: Real-time Botpress readiness badge reflecting cloud connectivity.
4. High-Impact Hero Section: Interactive SaaS preview card showcasing sample queries and pipeline.
5. Interactive Guidance Topic Search: Real-time client-side filter dynamically isolating support cards.
6. Six Core Support Categories: Academic Support, Exam Rules, Study Planning, Resources, Admissions, Campus Life.
7. Five Pre-Engineered Prompts: One-click interactive inquiry cards launching rich conversational sessions.
8. Curated Knowledge Base Suite: Modular markdown files covering degree criteria, grading scales, and library commons.
9. Official Botpress Webchat Integration: Direct embed of Webchat v3.7 with custom styling and event hooks.
10. Natural Language Understanding: Interprets colloquial syntax, typos, and shorthand gracefully.
11. Anti-Hallucination Guardrails: Strict refusal to guess official college schedules or fees.
12. Real-User Telemetry: Edge-level Core Web Vitals monitoring ensuring sub-second load times.

5. Technology Stack

Technology	Layer	Architectural Role & Justification
HTML5	Markup & Semantics	Semantic DOM landmarks, ARIA accessibility, WCAG 2.1 AA compliance.
CSS3	Styling & Layout	CSS Custom Properties, Flexbox, CSS Grid; zero utility framework overhead.
Vanilla JavaScript	Client Logic	DOM event binding, dynamic topic search filter,

		Botpress lifecycle sync.
Botpress Cloud	Conversational AI	Autonomous Node, NLU, dialogue management, LLM orchestration.
Botpress Knowledge Base	RAG Vector Store	Vector similarity retrieval over curated campus markdown documentation.
Botpress Webchat v3.7	Chat Interface	Slide-over drawer, typing indicators, secure HTTPS/WSS communication.
Git & GitHub	Version Control	Semantic commits, branch workflows, public open-source repository.
Edge Infrastructure	Hosting & Telemetry	SSL/TLS encryption, edge CDN caching, real-user Core Web Vitals tracking.

6. Project Screenshots (Proof of Work)

Screenshots represent mandatory proof of work verifying the authentic development and live production status of CampusConnect AI. In accordance with academic integrity guidelines, no synthetic mockups or AI-generated images are used. The 13 screenshots specified below document genuine application interactions across desktop, mobile, and cloud environments.

SCREENSHOT PROOF-OF-WORK: Figure 6.1 – Homepage and Navigation Overview

- **Expected Asset:** `screenshots/01-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for Homepage and Navigation Overview.
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.1: Demonstration of Homepage and Navigation Overview within the live CampusConnect AI production environment.

SCREENSHOT PROOF-OF-WORK: Figure 6.2 – Main Hero Section & SaaS Card Preview

- **Expected Asset:** `screenshots/02-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for Main Hero Section & SaaS Card Preview.
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.2: Demonstration of Main Hero Section & SaaS Card Preview within the live CampusConnect AI production environment.

SCREENSHOT PROOF-OF-WORK: Figure 6.3 – Platform Features Section

- **Expected Asset:** `screenshots/03-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for Platform Features Section.
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.3: Demonstration of Platform Features Section within the live CampusConnect AI production environment.

SCREENSHOT PROOF-OF-WORK: Figure 6.4 – Student Support Categories & Search Filter

- **Expected Asset:** `screenshots/04-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for Student Support Categories & Search Filter.
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.4: Demonstration of Student Support Categories & Search Filter within the live CampusConnect AI production environment.

SCREENSHOT PROOF-OF-WORK: Figure 6.5 – Actual Integrated Botpress Webchat Interface

- **Expected Asset:** `screenshots/05-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for Actual Integrated Botpress Webchat Interface.
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.5: Demonstration of Actual Integrated Botpress Webchat Interface within the live CampusConnect AI production environment.

SCREENSHOT PROOF-OF-WORK: Figure 6.6 – Real AI-Generated Conversational Response

- **Expected Asset:** `screenshots/06-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for Real AI-Generated Conversational Response.
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.6: Demonstration of Real AI-Generated Conversational Response within the live CampusConnect AI production environment.

SCREENSHOT PROOF-OF-WORK: Figure 6.7 – RAG Knowledge-Base-Supported Answering

- **Expected Asset:** `screenshots/07-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for RAG Knowledge-Base-Supported Answering.
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.7: Demonstration of RAG Knowledge-Base-Supported Answering within the live CampusConnect AI production environment.

SCREENSHOT PROOF-OF-WORK: Figure 6.8 – Study-Related Assistance Interaction

- **Expected Asset:** `screenshots/08-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for Study-Related Assistance Interaction.
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.8: Demonstration of Study-Related Assistance Interaction within the live CampusConnect AI production environment.

SCREENSHOT PROOF-OF-WORK: Figure 6.9 – Handling of Unknown Question (Anti-Hallucination Guardrail)

- **Expected Asset:** `screenshots/09-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for Handling of Unknown Question (Anti-Hallucination Guardrail).
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.9: Demonstration of Handling of Unknown Question (Anti-Hallucination Guardrail) within the live CampusConnect AI production environment.

SCREENSHOT PROOF-OF-WORK: Figure 6.10 – Botpress Cloud Knowledge Base Configuration

- **Expected Asset:** `screenshots/10-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for Botpress Cloud Knowledge Base Configuration.
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.10: Demonstration of Botpress Cloud Knowledge Base Configuration within the live CampusConnect AI production environment.

SCREENSHOT PROOF-OF-WORK: Figure 6.11 – Botpress Autonomous Node Configuration & Agent Prompt

- **Expected Asset:** `screenshots/11-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for Botpress Autonomous Node Configuration & Agent Prompt.
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.11: Demonstration of Botpress Autonomous Node Configuration & Agent Prompt within the live CampusConnect AI production environment.

SCREENSHOT PROOF-OF-WORK: Figure 6.12 – Mobile Responsive Layout

- **Expected Asset:** `screenshots/12-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for Mobile Responsive Layout.
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.12: Demonstration of Mobile Responsive Layout within the live CampusConnect AI production environment.

SCREENSHOT PROOF-OF-WORK: Figure 6.13 – Live Production Website Deployment

- **Expected Asset:** `screenshots/13-screenshot.png`
- **Viewport / Source:** Production / Cloud Studio | URL: https://campus-connect-kohl-alpha.vercel.app/
- **Required Evidence:** Authentic visual evidence for Live Production Website Deployment.
- **Capture Status:** **Documented Specification (Manual Browser Capture by Student Required Prior to Final Submission)**

Figure 6.13: Demonstration of Live Production Website Deployment within the live CampusConnect AI production environment.

7. Open Source Repository

The complete source code, architectural documentation, Botpress configuration, and knowledge base repository for CampusConnect AI are maintained publicly on GitHub:

Official GitHub Repository: https://github.com/Immortal-syntax/CampusConnect

- Repository Identifier: Immortal-syntax/CampusConnect
- Default Branch: main

• License: MIT Open Source License
• Hosting: GitHub Cloud Infrastructure

8. Future Scope & Conclusion

Realistic enhancements for future iterations include:
1. Multilingual Support: Regional vernacular language translation (Kannada, Hindi, Tamil, Telugu, Bengali).
2. Voice Interface: Speech-to-Text and Text-to-Speech capabilities for hands-free and accessible interaction.
3. Personalized Trajectories: Secure opt-in course enrollments to suggest customized weekly review timetables.
4. Progressive Web App: Service Worker caching for home-screen installability and offline emergency access.
5. Institutional ERP / SIS SSO: Authenticated queries for personal attendance and transcript credit audits.
6. Human Advisor Escalation: Intelligent routing of sensitive student crises to counseling staff.

Conclusion: CampusConnect AI demonstrates how Generative AI, when thoughtfully combined with standard web technologies and grounded in factual institutional knowledge, serves as an equitable digital assistant. By providing 24/7 academic advising, teaching evidence-based study methods, and connecting learners with verified open educational repositories, CampusConnect AI transforms the collegiate experience and makes a tangible contribution toward realizing United Nations Sustainable Development Goal 4: Quality Education.

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